

Nishant Kumar

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Education

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar

2022 – 2026

Bachelor of Technology

Technical Skills

- **Languages:** JavaScript, TypeScript, Python, Java
- **Frontend:** React.js, Redux, HTML, CSS, Tailwind CSS
- **Backend:** Node.js, Express.js, REST APIs, WebSockets, JWT, Argon2/bcrypt
- **Databases:** PostgreSQL, Prisma, Redis, MongoDB
- **Cloud & DevOps:** AWS (EC2, S3), Docker, CI/CD
- **Tools:** Git, GitHub, Postman, BullMQ

Projects

Snap Link - a Url shortner

[GitHub](#)

TypeScript, Node.js, Express, PostgreSQL, Prisma, Redis, BullMQ, JWT

- Engineered a scalable URL shortening backend following a layered Controller–Service–Repository architecture with TypeScript, Express, PostgreSQL, and Prisma.
- Secured user accounts with JWT access/refresh-token rotation, Argon2 password hashing, email verification, and password-reset flows.
- Integrated Redis caching and BullMQ background jobs to process URL analytics asynchronously, reducing load and improving latency on the redirect path.
- Added custom aliases, link expiration, protected URLs, and click-tracking analytics, backed by Zod validation and centralized error handling.
- Applied database indexing on frequently queried fields to keep redirect and lookup queries performant as the dataset grows.
- Structured the codebase with clear separation of concerns across controllers, services, and repositories for long-term maintainability.

We Draw - Collaborative Whiteboard

[GitHub](#)

React, TypeScript, Canvas API, WebSocket, AWS

- Developed a real-time collaborative whiteboard enabling multiple users to draw and edit on a shared canvas simultaneously.
- Crafted freehand drawing, shape creation, and drag-and-drop interactions using HTML Canvas and TypeScript.
- Designed a WebSocket-based synchronization layer to broadcast drawing events and keep canvas state consistent across clients.
- Optimized rendering with requestAnimationFrame, React refs, and selective canvas redraws to minimize unnecessary re-paints.
- Enabled element selection and manipulation (move, resize, delete) so users can edit existing shapes, not just draw new ones.
- Deployed the frontend on AWS S3 static hosting and configured the WebSocket backend for reliable end-to-end connectivity.

Splitzy – Group Expense Manager

[GitHub](#)

React, TypeScript, Node.js, Express, PostgreSQL, Prisma, JWT

- Architected a full-stack expense management app for creating groups, tracking shared expenses, and managing settlements.
- Implemented 10+ REST APIs using Node.js, Express, and Prisma ORM across user, group, expense, and settlement workflows.
- Protected routes with JWT-based authentication and role-based authorization.
- Modeled a normalized PostgreSQL schema using Prisma to efficiently manage users, groups, expenses, and settlements.
- Devised a balance-settlement algorithm to calculate who owes whom within a group and minimize the number of transactions needed to settle up.
- Delivered a responsive React and Tailwind CSS frontend, ensuring a consistent experience across mobile and desktop screen sizes.

Achievements

- Strong grasp of Data Structures and Algorithms including DP, Graphs, and Recursion, solving 300+ DSA problems on [LeetCode](#),gfg etc.